

Mobile Communications' Base Stations and Wi-Fi Access Points inside a Cruise Ship

Group 08

Alice Navalho · Beatriz Coelho · Luís Nóbrega

Maria Abrantes · Mónica Ferreira · Rodrigo Esgueira

Instituto Superior Técnico

Outline

- Understanding the Onboard Challenge
- Scope, Assumptions and Requirements
- Stakeholder
- Work Breakdown Structure
- RACI Matrix
- Resources & Cost Analysis
- Schedule
- Quality Control Plan
- Risk Management
- Conclusions

Understanding the Onboard Challenge

OUR SHIP: MV WORLD VOYAGER

200

Passengers

120

Crew members

6

Main decks



KEY CONNECTIVITY CHALLENGES

Metallic Structure

Compartmentalized
Layout

Satellite Backhaul
at Sea

Port Shutdown
Requirement

Scope, Assumptions & Requirements

Scope

Radio-frequency Planning

Base Stations

Port Shutdown

Antennas

Wi-Fi

Backhaul Integration

Assumptions and Constraints

Deck Documentation

Satellite Provider

Regulation

80; 100; 120 €/h

Shutdown

Deadlines

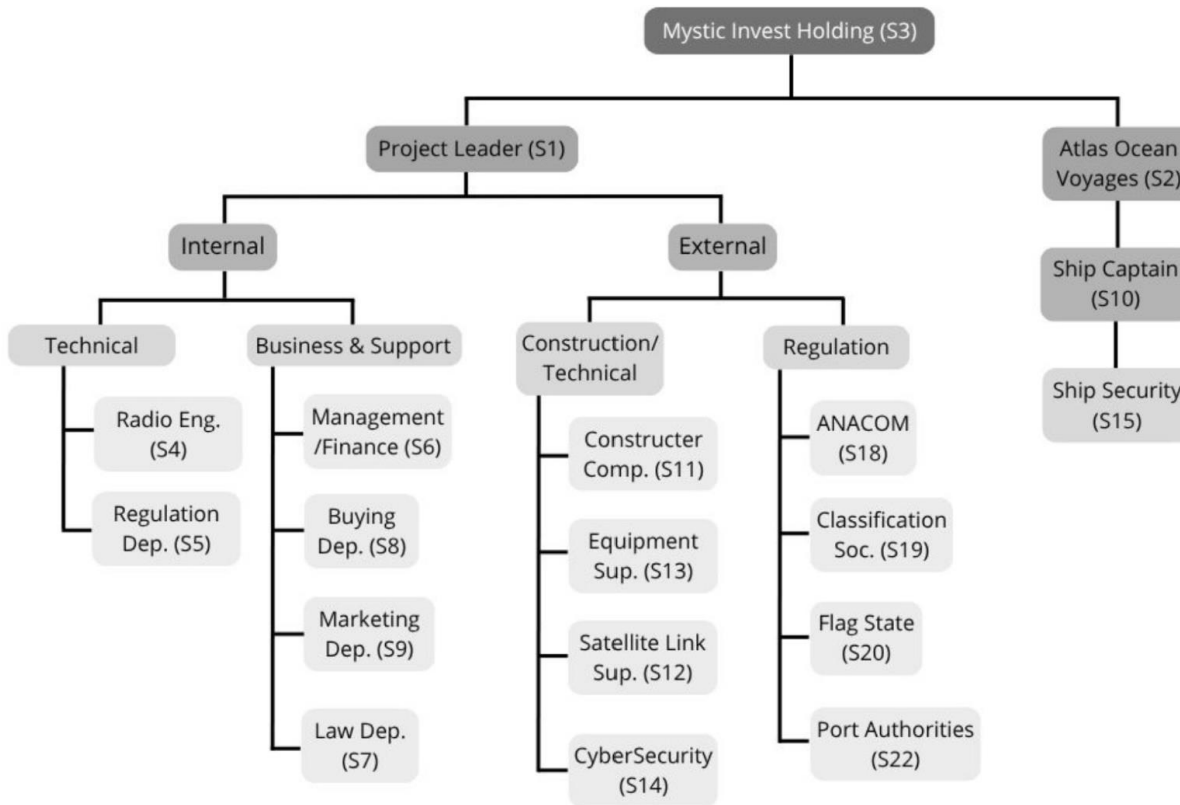
Aesthetic

Requirement

Business

Technical

Project Management



24 stakeholders

7 internal

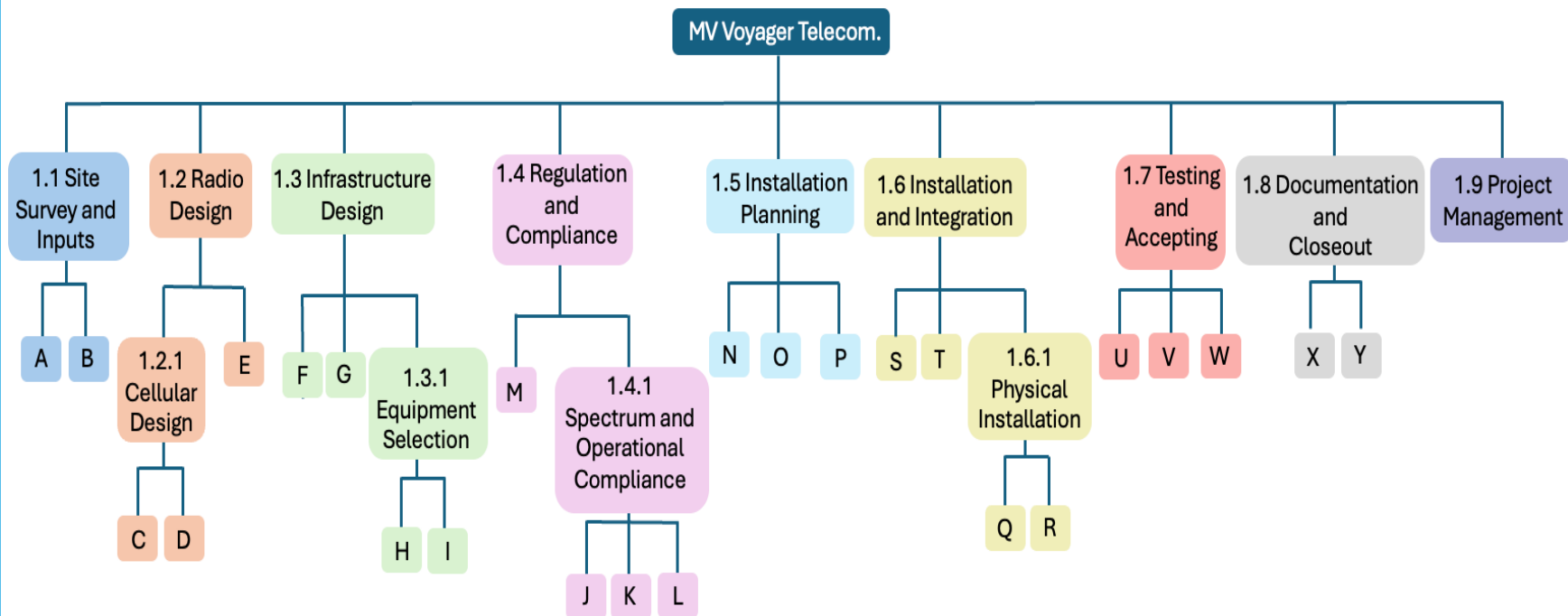
17 external

8 Positive

15 Mixed/Neutral

1 Negative

Work Breakdown Structure



RACI Matrix

Code	Activity	Project Leader	Radio. Eng	Reg. Dept	Bus. & Sup	Constr (Ext)	Other
E	Wi-Fi AP Layout & Service Zone Design	A	R	C	C	C	Cybersecurity
L	Regulatory Conformity Analysis	A	C	R	C		Reg. Agencies
Q	Install Cabling, Power, Base Stations & Antennas	A	C			R	Capt. & Sec.
Y	Handover & Sign-off	R&A	R		I		Client

A Accountable
 R Responsible
 C Consulted
 I Informed
 R&A Combined

Resources & Cost Analysis

120 €/hr

- Project Leader
- Radio Eng. Sr.
- Law Officer

100 €/hr

- Radio Eng. (2)
- Regulation Officer
- Finance/Mngm. Analyst

80 €/hr

- Buying Officer
- Marketing Specialist

External

- Constructor Fee (5): 100€/hr
- Cybersec. Consult. Fee: 120€/hr

Material

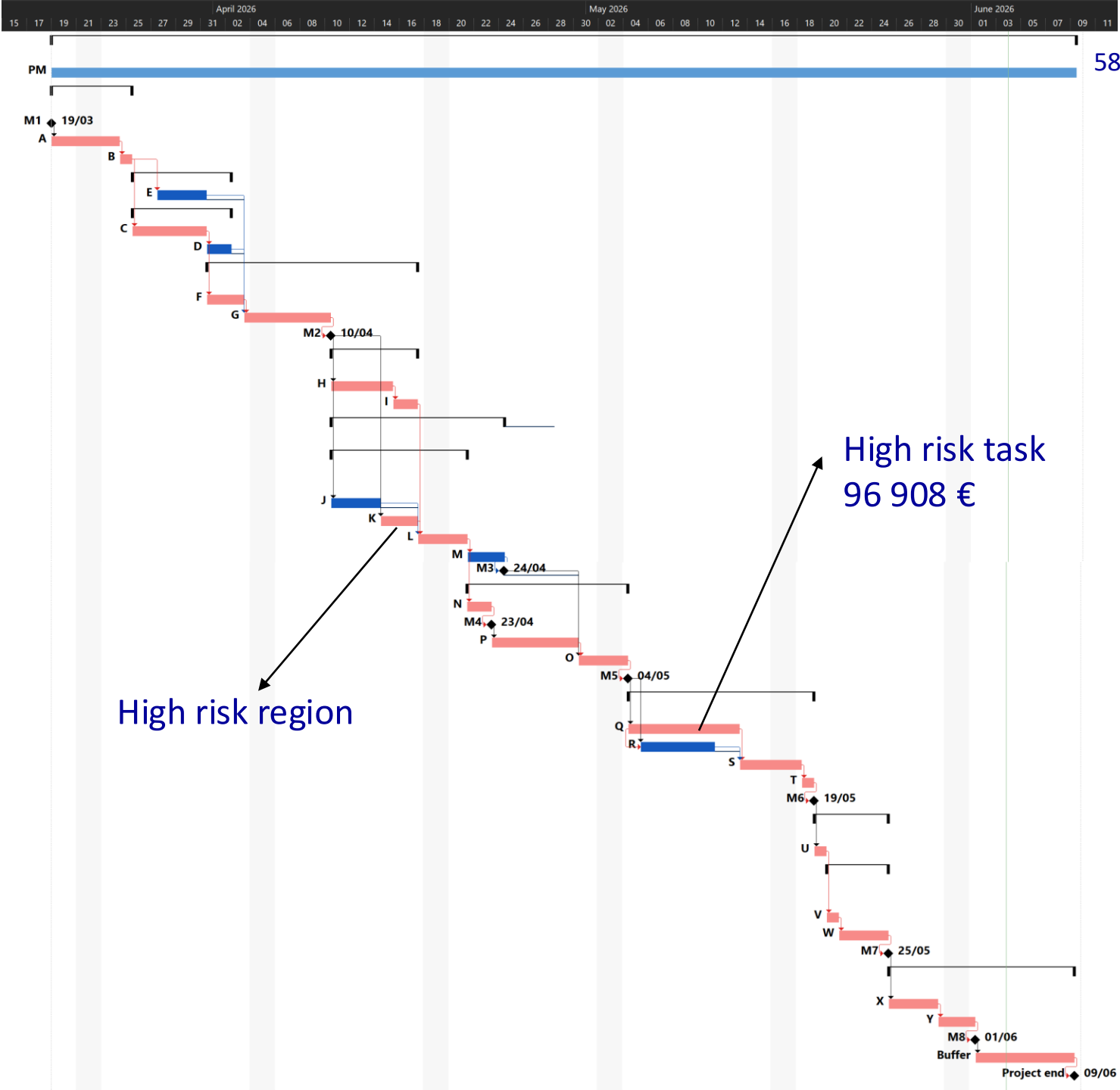
- Regulatory & License Fees: 3 500€
- Outdoor Antenna: 30 000€
- Satellite Sign-Up Fee: 5 000€



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LISBOA

Schedule

GPE



58,5 days

High risk region

High risk task
96 908 €

Project end, 09/06

Quality Control Plan

Checkpoints

- Design review
- Procurement review
- Testing review
- Final acceptance

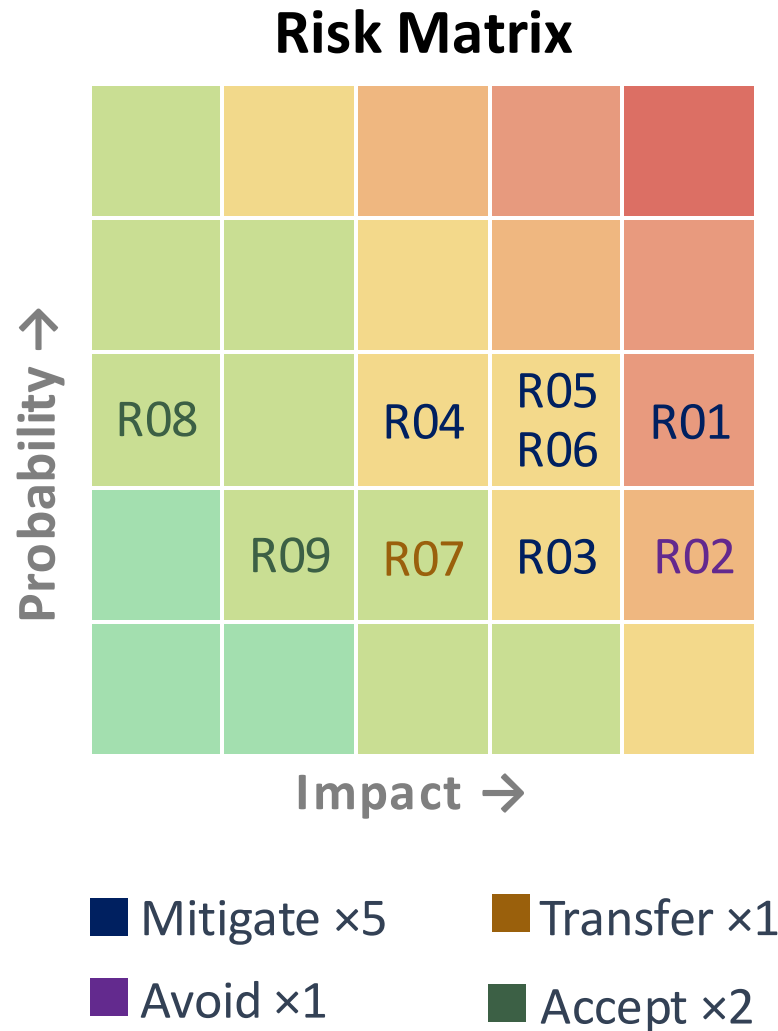
Main criteria

- 95% coverage
- 325 users + 20% headroom
- Latency < 700 ms
- Shutdown < 30 s

Control logic

- Verification method
- Responsible owner
- Corrective action
- Bi-weekly EVM review

Risk Management



Contingency Reserve

15 174 €

≈ 7% of the 213 996 € estimate

Expected Delay

≈ 5,5 d

Within the 6 days buffer

Conclusions

Complete and realistic planning package for the MV World Voyager telecommunications deployment.



WBS with clear deliverable traceability.
PERT and resource leveling supported execution planning.

Risk and quality increased budget realism.
Compliance and logistics matched the marine context.

Structured, traceable, and implementation-ready.